ENTERPRISE IT APPLICATION ARCHITECTURE SPECIFICATION

InfraPilot: End-to-End Workflow & Orchestration Guide

Detailed Technical Specification of Request Lifecycles, Quota Gating, BullMQ Orchestration, Terraform Sandboxing, and Fleet Governance

1. Enterprise Workflow Overview & Architecture Topology

InfraPilot is an enterprise self-service infrastructure management platform that orchestrates the complete lifecycle of virtual machine provisioning. The platform combines an Express.js REST API with an autonomous BullMQ background worker powered by Redis and HashiCorp Terraform. Operations are completely decoupled: HTTP web requests never block on infrastructure execution, eliminating 504 timeouts and ensuring maximum reliability.

2. End-to-End Sequence & Swimlane Workflow

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1. DEV: Submit VM Request (CPU, RAM, Disk, OS) -> 2. REST API: Validate schema & enforce quota
3. API: Status -> PENDING_APPROVAL -> 4. ADMIN: Review queue -> Click 'Approve'
5. API: Atomic Transaction (Status -> APPROVED, Insert Approval, Create Job, Log Audit)
6. API: Enqueue Job to BullMQ ('provisioning' queue in Redis) -> Return 200 OK to UI
7. WORKER: Atomically pops job -> Updates DB Stage: VALIDATING -> Runs terraform validate
8. WORKER: Updates DB Stage: PLANNING -> Runs terraform plan
9. WORKER: Updates DB Stage: APPLYING -> Runs terraform apply (Local Provider sandbox)
10. WORKER: Atomic Transaction (Create VirtualMachine ACTIVE, Job -> COMPLETED, Audit Log)
11. DEV/ADMIN: Dashboard shows Active VM with deterministic IP & interactive power controls
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3. Request Initiation & Quota Gating Workflow

Requests pass through two strict validation tiers before admission into the workflow:

• **Tier 1: Zod Schema Boundaries**: Enforces CPU (1-64), RAM (1024-262144 MB), Disk (10-4096 GB), OS whitelist, and justification text.

• **Tier 2: Real-Time Server-Side Quota Gate**: Evaluates (Committed_Resources + Requested_Resources <= Quota_Limit). If exceeded, the request is immediately rejected with HTTP 400 and excess calculations, preventing database pollution.

4. The 8-State Finite State Machine (FSM) Lifecycle

Source State	Allowed Targets	Actor	Trigger Event & Side Effect
DRAFT	PENDING_APPROVAL, CANCELLED	Developer	Developer submits or cancels draft. Quota evaluated.
PENDING_APPROVAL	APPROVED, REJECTED, CANCELLED	Admin / Dev	Admin approval/rejection or developer cancellation. On APPROVE, job created.
APPROVED	PROVISIONING, FAILED	Worker	Worker pops job from BullMQ. Status moves to PROVISIONING.
PROVISIONING	ACTIVE, FAILED	Terraform	Terraform apply completes -> ACTIVE. Execution error -> FAILED.
ACTIVE	DECOMMISSIONED	Dev / Admin	Instance decommissioned. Quotas freed in real-time.
FAILED	PROVISIONING, CANCELLED	Admin	Manual retry re-queues job to BullMQ. Cancellation frees quota.
REJECTED	Terminal State	System	Terminal audit record. Reserved quota commitment is released.

5. Governance & Administrative Approval Workflow

• **Separation of Duties (RBAC)**: Developers are strictly barred from approving requests (HTTP 403 Forbidden).
• **Anti-Self-Approval**: Administrators cannot approve requests they personally authored.
• **Double-Approval Prevention**: Database row-level locks prevent race conditions; second approvals return HTTP 400.
• **Mandatory Audit Trail**: Every decision requires an administrative comment recorded in the Approval history table.

6. BullMQ Asynchronous Orchestration & Terraform Sandboxing

• **BullMQ Queueing**: Jobs are queued in Redis with 3 retry attempts and exponential backoff.
• **Sandbox Directory Isolation**: The worker creates isolated workspaces (/runs/<jobId>) per execution.
• **4-Stage Pipeline**: (1) Workspace Init -> (2) terraform validate -> (3) terraform plan -> (4) terraform apply.
• **Security Controls**: Pinned binary invocation (/usr/local/bin/terraform), shell: false argument arrays (no injection), 60s hard timeout.
• **Local Provider Output**: Writes JSON manifests (/manifests/vm-*.json) containing assigned hostname, IP (10.0.X.Y), and MAC address without incurring cloud bills.

7. Fleet Inventory, Power States, & Safe Failure Recovery

• **Inventory Lifecycle**: Active VMs are bound to their owner and request. Power states (RUNNING / STOPPED) can be toggled interactively.
• **Zero Ghost Inventory Guarantee**: If Terraform fails at any stage, VM creation is aborted, leaving no orphan records.
• **Diagnostic Logs & Safe Retries**: Full stdout/stderr streams are saved. Administrators can trigger safe retries directly from the UI.
• **Immutable Compliance Audit Trail**: All logins, requests, approvals, job retries, and power actions are stored in an append-only audit ledger.

8. Operator Credentials & Verification Flow

Persona	Email	Password	Access Scope
Enterprise Admin	admin@infrapilot.local	AdminPass123!	Global Fleet, Approvals, Quotas, Audit Logs, Retries
Senior Developer	developer@infrapilot.local	DevPass123!	Self-Service Requests, Personal Inventory, Power State
DevOps Engineer	developer2@infrapilot.local	DevPass123!	Tenant Isolation test (isolated inventory)

Verification Steps in Browser: Open **http://localhost:5173** -> Log in as Developer -> Submit VM request -> Log in as Admin -> Approve in Approvals -> Watch Provisioning Jobs live progress -> View VM Inventory and toggle power.